



First Semester: 2026-2027
Course Handout: Part-II

Date: 31-07-2026

In addition to Part-I (General handout for all courses appended to the timetable) this portion gives further specific details regarding the course:

Course No.	: CS U425
Course Title	: Deep Learning
Instructors	: Dr. Bharat Richhariya [IC] [bharat.richhariya@pilani.bits-pilani.ac.in] : Ghufuran Alam Siddiqui (RS) [p20230434@pilani.bits-pilani.ac.in]

I. Objective and Scope of the Course

In the last decade, due to the availability of cheap computational power and relevant software tools, several neural networks are explored to advance state-of-the-art of many hard AI problems. These architectures have successfully advanced various areas such as image searching, natural language understanding, medical applications, autonomous vehicles etc. All these problems rely on efficient, accurate and robust solutions for basic vision tasks like image classification, localization and detection. In this course students will be given an exposure to the details of neural networks as well as deep learning architectures and to develop end-to-end models for such tasks. Students will learn to implement, train and debug their own neural networks. This is a project oriented practical course in which every student has to develop a complete working model to solve some real world problem.

2. Course Material

Text Book:

[TB1]:	Ian Goodfellow, Yoshua Bengio and Aaron Courville, Deep learning, In preparation for MIT Press, Available online: http://www.deeplearningbook.org , 2016
[TB2]:	Aston Zhang and Zachary C. Lipton and Mu Li and Alexander J. Smola, Dive into Deep Learning, https://d2l.ai/index.html 2021

Reference Books:

[R1]:	Satish Kumar , Neural Networks - A Class Room Approach, Second Edition, Tata McGraw-Hill, 2013
[R2]:	S. Haykin , Neural Networks and Learning Machines, Prentice Hall of India, 2010
[R3]:	Kevin Murphy , Machine Learning: A Probabilistic Perspective, MIT Press, 2012 Online Link https://mitpress.mit.edu/books/machine-learning-0
[R4]:	Christopher M. Bishop , Pattern Recognition and Machine Learning, Springer, 2006.



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3. Course Plan

Lecture	Topic(s) to be discussed	Reference	Learning Objective
1-5	Introduction to machine learning, end to end learning, optimization, linear model of regression and its convergence, loss function, function approximation, evaluation parameters	TB1, R1, R3, R4 and lecture notes	Learning the power of linear models and general ML pipeline
6-8	Perceptron, multi-layer perceptron, activation functions, neuron, computational models of neurons, artificial neural networks (ANN), multilayer feedforward neural networks (MLFFNNs)	TB1, TB2, R1, R2 and lecture notes	Understand Model hierarchy and non linearity
9-13	Training feedforward neural networks, gradient descent, backpropagation learning, Regularization methods (L1, L2, dropout, batch normalization), loss landscape, parameter initialization.	TB1, TB2, R2 and lecture notes	Ability to train deep neural networks
14-16	Deep neural networks (DNNs), difficulty of training DNNs, Greedy layer wise training, Optimization for training DNNs, optimization methods for neural networks (SGD, AdaGrad, RMSProp, Adam).	TB1, TB2 and lecture notes	Design a trainable and suitable NN for a learning problem
17-20	Convolution neural networks (CNNs), convolution, pooling and its variations, striding, CNN training and fine-tuning: weights initialization, hyper-parameter tuning. Different deep CNN architectures - LeNet, AlexNet, VGGnet.	TB1, TB2 and lecture notes	Ability to design computer vision applications
21-23	Understanding and visualizing CNNs. Image classification with CNNs. Applications in Computer Vision.	TB1, TB2 and lecture notes	Train DNNs for computer vision
24-28	Recurrent neural networks (RNNs): Sequence modeling using RNNs, Back propagation through time, Vanishing and exploding gradients, Long short term memory (LSTM), Bidirectional LSTMs/RNNs, GRU	TB1, TB2 and lecture notes	Design DNN for text/ voice/ video or sequence data
29-33	Generative models, Autoencoders, Variational Autoencoders (VAE), Generative adversarial networks (GAN), encoder decoder architecture	TB1, TB2 and lecture notes	Apply generative approach to solve problems
34-37	Advanced Topics in Deep Learning: Interpretability, Explainability, Attention, Transformers, Large Language Models (LLMs)	Lecture notes and research papers	Understand the recent advances in DL
38-40	Applications: Applications in vision, speech and natural language processing.	Lecture notes and research papers	Appreciate real-life applications of DL

4. Learning outcome

Students who complete this course would be able

1. To understand importance of learning when sufficient data and processing power is available.
2. To understand the importance of general purpose solution with end-to-end learning.



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3. To understand the hierarchical training and learning of features.
4. To apply deep learning for basic computer vision tasks such as image classification, localization and detection.
5. To apply deep learning for generating interesting and new patterns using GAN.
6. To understand how to apply deep learning in sequence discovery problems such as text, voice, video etc.

5. Evaluation Scheme

S.No.	Evaluation Component	Duration	Weightage	Component type	Date and Time
1.	Mid-Semester Test	90 min	25%	CB	07/10 FN1
2.	Term Project	Take-home	20%	OB	TBA
3.	Lab assessment	-	10%	OB	Weekly
4.	Quiz*	50 mins	10%	CB	TBA
5.	Comprehensive Exam	3 hours	35%	CB or Partially OB	8/12 FN

* For quiz, a total of two quizzes will be conducted, and the best of the two quizzes will be used for the course total. **THEREFORE, MAKE-UP QUIZ WILL NOT BE GRANTED.**

6. Honor Code

Plagiarism in any form shall NOT be tolerated (we would be using appropriate plagiarism detection tools). Student shall be awarded ZERO marks and case may be reported to the appropriate committee of the Institute for appropriate action. Every component is individual until specifically specified.

7. Notices

All notices would be put on NALANDA.

8. Make-up Policy

Make-up: To be granted only in case of serious illness or emergency. Please see AUGSD guidelines.

Make-up in Lab component: No makeup will be given.

Make-up in Quiz component: No makeup will be given.

9. Chamber Consultation Hours

Friday 04.00 – 5.00 PM. Seek an appointment by emailing the instructor.

Instructor-in-Charge



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